# Lingzhi Zhao

Ph.D. Candidate in Computer Science | University of Illinois Urbana-Champaign
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## RESEARCH INTERESTS

My research broadly focuses on **networked multimedia systems (video on demand, live streaming, XR)** and **intelligent mobile systems**. I am interested in designing efficient systems for **emerging human-centric media**, including XR, egocentric AI assistants, and intelligent video streaming and understanding.

## EDUCATION

**University of Illinois Urbana-Champaign** 2022 – 2026 (expected)
*Ph.D. in Computer Science · Advisor: Prof. Klara Nahrstedt*

**Shanghai Jiao Tong University** 2019 – 2022
*M.S. in Electrical and Computer Engineering · Advisor: Prof. Ying Cui*

**Shanghai University** 2015 – 2019
*B.S. in Electrical and Computer Engineering*

## SKILLS

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|---|---|
| **Tools and Languages** | Python, C++, MATLAB, Swift, Java, C#, Rust |
| **(Video) Systems & Dev.** | FFmpeg, DASH, NVCodec, Wireshark, iperf3, tc, Git, NS-3 |
| **Machine Learning** | PyTorch, Tokenizers, Transformers, Instruction Tuning |

## RESEARCH EXPERIENCE

**Multimedia Operating Systems and Networking (MONET) Lab, UIUC** Aug. 2022 – Present
*Graduate Research Assistant · Advisor: Prof. Klara Nahrstedt*

**GenStream: Generative Codec-based Real-time Video Streaming**

- Optimized a generative video compression model (Cosmos Tokenizer) on a single RTX 4090 GPU for real-time streaming, achieving 50%–78% bandwidth savings over H.264/H.265 and prior neural codecs at comparable quality.
- Proposed a practical token adaptation approach by exploiting video context diversity to handle network dynamics.
- Improved perceptual quality (LPIPS) by 38%–59% at matched bitrates and loss patterns compared to state-of-the-art streaming systems.

**AquaScope: Generative Codec-based Image Transmission on Mobile Systems** *[ACM MobiSys 2026] [Website]*

- Developed the first acoustic-based mobile system enabling reliable image transmission between mobile devices via generative compression (TiTok, VQGAN) in resource-constrained environments.
- Improved bandwidth efficiency by 40% and transmission error resilience via token clustering and fine-tuning, outperforming PNG, JPEG, and classic neural compression; reduced bit error rate (BER) from 19% to below 2% on average.

**AquaVLM: VLM-powered Intelligent Mobile Communication Systems** *[arXiv 2025] [Website]*

- Developed the first VLM-based mobile system for cooperative communication and situation awareness in extreme underwater environments.
- Instruction-tuned MobileVLM-V2 on multimodal data (underwater images + sensor readings) with a hierarchical message generation approach to align user intent while reducing inference latency.
- Built an iOS prototype with on-device inference (LLMFarm) and a chirp-based physical-layer protocol; constructed an end-to-end VR simulation platform for realistic evaluation.

**Multimedia Lab, ByteDance Inc., San Diego, CA** May 2024 – Dec. 2024 | May 2025 – Aug. 2025
*Research Intern · Manager: Dr. Shu Shi*

**MARS: Motion-Aware ABR for Real-Time VR** *[Under review, ACM Multimedia 2026]*

- Developed a perceptually-guided bitrate adaptation algorithm for a real-time VR streaming system, achieving 15% bandwidth reduction without degrading QoE, supported by a user study with real-world VR applications.
- Designed a kinematics-based state machine that synchronizes bitrate adaptation with head motion phases.
- Extended the system to support multiple rate control modes (CBR, VBR, CQP) and integrated motion vector extraction for rate adaptation optimization.

**Trinity: QoE Optimization for Cloud VR Gaming** *[ACM MMSys 2026] [Website]*

- Validated that users exhibit different latency tolerance across action types (head, hand, body motion) using an open-source first-person shooter VR game.
- Developed a delayed-rendering strategy for hand motion to bridge latency gaps and a customized rendering approach to generate partial panoramas for head motion.
- Prototyped an end-to-end cloud VR gaming system, achieving comparable QoE to local streaming systems while reducing bandwidth cost and GPU utilization by 60% and 40%, respectively.

**Institute of Wireless Communications Technology, Shanghai Jiao Tong University** Aug. 2020 – Mar. 2022

*Graduate Research Assistant · Advisor: Prof. Ying Cui*

**RL-based Adaptive Wireless Video Streaming** *[IEEE Trans. Multimedia 2025]*

- Theoretically analyzed and optimized a reinforcement learning (RL)-based adaptive bitrate (ABR) algorithm for video streaming, improving QoE by 14.4% and accelerating training convergence.
- Incorporated and mathematically modeled physical-layer network information (PRB count, MCS) into the RL framework to further boost QoE.
- Designed an online tuning approach via progressive neural networks to balance generalizability and environment-specific adaptation across diverse network conditions.
- Deployed the optimized algorithm in Tencent's production video streaming system.

## SELECTED PUBLICATIONS

**[1]** **Lingzhi Zhao**, Yongqiang Gui, Yanyan Suo, Sandesh Dhawaskar Sathyanarayana, Mingyuan Wu, Bo Chen, Ruixiao Zhang, Klara Nahrstedt, and Shu Shi. MARS: Streaming How You Move with Motion-Aware ABR for Real-Time VR. *ACM Multimedia 2026, under review.*

**[2]** Beitong Tian*, **Lingzhi Zhao***, Bo Chen, Mingyuan Wu, Haozhen Zheng, Deepak Vasisht, Francis Y. Yan, and Klara Nahrstedt. AquaScope: Reliable Underwater Image Transmission on Mobile Devices. (*: equal contributions) *ACM MobiSys 2026.*

**[3]** **Lingzhi Zhao**, Yongqiang Gui, Yanyan Suo, Sandesh Dhawaskar Sathyanarayana, Ruixiao Zhang, Shu Shi, and Klara Nahrstedt. Trinity: Exploiting Latency Sensitivity to Improve Quality of Experience on Cloud VR Gaming. *ACM MMSys 2026.*

**[4]** Beitong Tian*, **Lingzhi Zhao***, Bo Chen, Haozhen Zheng, Jingcheng Yang, Mingyuan Wu, Deepak Vasisht, and Klara Nahrstedt. AquaVLM: Improving Underwater Situation Awareness with Mobile Vision Language Models. (*: equal contributions) *arXiv preprint, 2025.*

**[5]** **Lingzhi Zhao**, Qian Zhou, Bo Chen, and Klara Nahrstedt. 360LiveCast: A Low-Latency and Bandwidth-efficient Multicast Framework for Live 360 Video. *IEEE MIPR 2025.*

**[6]** **Lingzhi Zhao**, Ying Cui, Yuhang Jia, Yunfei Zhang, and Klara Nahrstedt. Enhancing Neural Adaptive Wireless Video Streaming via Lower-Layer Information Exposure and Online Tuning. *IEEE Transactions on Multimedia 2025.*

**[7]** **Lingzhi Zhao**, Ying Cui, Sheng Yang, and Shlomo Shamai (Shitz). An Optimization Framework for General Rate Splitting for General Multicast. *IEEE Transactions on Wireless Communications 2022.*

**[8]** **Lingzhi Zhao**, Ying Cui, Zhi Liu, Yunfei Zhang, and Sheng Yang. Adaptive Streaming of 360 Videos with Perfect, Imperfect, and Unknown FoV Viewing Probabilities in Wireless Networks. *IEEE Transactions on Image Processing 2021.*

## THESIS COMMITTEE

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|---|---|
| **Klara Nahrstedt** *(Chair)* | University of Illinois Urbana-Champaign |
| **Francis Y. Yan** | University of Illinois Urbana-Champaign |
| **Philip Brighten Godfrey** | University of Illinois Urbana-Champaign |
| **Shu Shi** | ByteDance |